Summary Notes:
AWS Databases

1. Introduction

Overview

Here is an overview of the topics covered in the session:

1. Overview of AWS Databases
    - Definition and basic concepts
    - Key characteristics of AWS Databases
    - Types of AWS database services
    - Advantages and disadvantages of AWS Databases
2. Amazon RDS
    - Introduction to Amazon RDS
    - Key Features
    - Supported Database Engines
    - Pricing and Cost Management
    - Security and Compliance
    - Performance and Scalability
    - Management and Administration
    - Backup and Recovery
    - Migration and Integration
    - Best Practices
3. Amazon Aurora
    - Introduction to Amazon Aurora
    - Key Features
    - Performance and Scalability
    - High Availability and Durability
    - Supported Database Engines
    - Backup and Recovery
    - Best Practices
4. Amazon Redshift
    - Introduction to Amazon Redshift
    - Key Features
    - Performance and Scalability

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- High Availability and Durability
- Data Warehousing Architecture
- Query and Data Management
- Integration With BI Tools
- Backup and Recovery
- Best Practices

5. Amazon Neptune
    - Introduction to Amazon Neptune
    - Key Features
    - Performance and Scalability
    - High Availability and Durability
    - Graph Database Architecture
    - Query and Data Management
    - Integration With Analytics and Visualisation Tools
    - Backup and Recovery
    - Best Practices
6. Amazon DynamoDB
    - Introduction to Amazon DynamoDB
    - Key Features
    - Performance and Scalability
    - Data Modeling
    - NoSQL Database Concepts
    - Consistency Models
    - Global Tables and Multi-Region Replication
    - Backup and Recovery
    - Integration With Other AWS Services
    - Best Practices
7. Amazon DocumentDB
    - Introduction to Amazon DocumentDB
    - Key Features
    - Performance and Scalability
    - High Availability and Durability
    - MongoDB Compatibility
    - Query and Indexing
    - Backup and Recovery
    - Best Practices
8. Hands-on Lab/Demo
    - How to Build Your DB Server
    - How to Interact With Your DB Using an App

© Copyright upGrad Education Pvt. Ltd. All rights reserved

2. Overview of AWS Databases

2.1 AWS Databases

Amazon Web Services (AWS) offers a wide range of database services designed to meet the diverse needs of modern applications. These managed services provide scalable, secure and highly available database solutions that reduce the operational complexity of database management. AWS databases cater to various data models and workloads, from relational and key-value to documents, graphs and in-memory databases.

AWS databases simplify administrative tasks, enhance scalability and ensure high availability, making it easier for businesses to manage their data efficiently.

2.2 Key Characteristics

Here are the key characteristics of AWS databases:

- **Managed services**: Automate administrative tasks such as backups, patching and scaling, reducing the burden on IT teams
- **Scalability**: Easily adjust database resources to meet changing demands, with options for both vertical and horizontal scaling
- **High performance**: Optimised for fast and efficient data processing, supporting high transaction rates and complex queries
- **Security**: Offer comprehensive security features, including encryption at rest and in transit, network isolation via VPC and IAM for access control
- **High availability and durability**: Involve built-in redundancy and failover mechanisms, with multi-AZ deployments and automated backups
- **Integration with the AWS ecosystem**: Seamlessly integrate with other AWS services such as Lambda, S3 and CloudWatch
- **Cost management**: Offer flexible pricing models, including pay-as-you-go and reserved instances, along with cost optimisation tools

© Copyright upGrad Education Pvt. Ltd. All rights reserved

2.3 Types of AWS Database Services

- **Amazon RDS**
  - A fully managed relational database service that supports multiple database engines (MySQL, PostgreSQL, Oracle and SQL Server)
- **Amazon Aurora**
  - A high-performance, relational database service that is compatible with MySQL and PostgreSQL with automatic scaling and high availability
- **Amazon Redshift**
  - A fully managed data warehouse service optimised for large-scale data analysis and complex queries
- **Amazon Neptune**
  - A managed graph database service used to build and run applications that work with highly connected data sets
- **Amazon DynamoDB**
  - A managed NoSQL database service offering key-value and document data structures with single-digit millisecond performance
- **Amazon DocumentDB**
  - A fully managed document database service designed for scalability and high availability that is compatible with MongoDB workloads

2.4 Advantages and Disadvantages

Given below are the advantages of AWS databases:

- **Managed services**: Automate administrative tasks such as backups, patching and scaling, reducing the burden on IT teams
- **Scalability**: Easily scale database resources up or down to meet changing workload demands, both vertically and horizontally
- **High performance**: Optimised for fast and efficient data processing, supporting high transaction rates and complex queries
- **Security**: Offer comprehensive security features, including encryption at rest and in transit, network isolation via VPC and IAM for fine-grained access control
- **High availability and durability**: Have built-in redundancy and failover mechanisms, with multi-AZ deployments and automated backups ensuring uptime and data protection
- **Integration with the AWS ecosystem**: Seamlessly integrate with Lambda, S3, CloudWatch and other AWS services for enhanced functionality
- **Cost management**: Offer flexible pricing models, including pay-as-you-go and reserved instances, along with cost optimisation tools
- **Global reach**: AWS's global infrastructure allows for data replication and disaster recovery across different regions and availability zones

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Innovation and continuous improvement**: Regular updates and new features keep AWS databases at the forefront of technology

These are the disadvantages of AWS databases:

- **Cost management complexity**: While managing costs is flexible, it effectively requires careful monitoring and optimisation to avoid unexpected charges.
- **Vendor lock-in**: Dependence on AWS services and infrastructure can make it challenging to migrate to other platforms or providers.
- **Learning curve**: Effectively learning and managing various AWS database services requires time and expertise.
- **Limited customisation**: Managed services may offer less flexibility for customisation compared with self-managed database solutions.
- **Performance variability**: Performance can sometimes vary based on shared infrastructure and network latency, especially during peak usage times.
- **Compliance and regulatory concerns**: Ensuring compliance with various regulatory standards may require additional configurations and monitoring.
- **Data transfer costs**: Moving data in and out of AWS services can incur additional costs, particularly for large data sets or frequent data transfers.

3. Amazon RDS

3.1 Introduction to Amazon RDS

Amazon RDS is a managed database service that takes care of most database management tasks, allowing you to focus on your application and your users. Amazon RDS helps save time and resources by eliminating tedious manual tasks such as hardware provisioning, database setup, patching and backups. It supports multiple database engines, including MySQL, PostgreSQL, Oracle, MariaDB and Microsoft SQL Server, giving you the flexibility to choose the best database for your specific needs.

3.2 Key Features

Here are the key features of Amazon RDS:

- **Automated backups:** Regular backups are automatically created and stored, ensuring data durability and facilitating point-in-time recovery.
- **Multi-AZ deployment:** High availability is achieved through automated failover across multiple Availability Zones.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Read replicas:** Amazon RDS improves the read performance and scale-out workloads by creating up to five read replicas.
- **Automatic software patching:** It keeps your database instances up-to-date with the latest patches and updates.
- **Monitoring and metrics:** It integrates with Amazon CloudWatch to monitor database performance and set alarms.

3.3 Supported Database Engines

These are the database engines supported by Amazon RDS:

- **MySQL**: A popular open-source relational database known for its reliability and ease of use
- **PostgreSQL**: An advanced open-source relational database renowned for its feature richness and standards compliance
- **MariaDB**: A community-developed fork of MySQL, offering additional features and improved performance
- **Oracle**: A commercial database known for robust performance, extensive features and enterprise-grade capabilities
- **Microsoft SQL Server**: A comprehensive, enterprise-class relational database from Microsoft, widely used in business applications

3.4 Pricing and Cost Management

Given below are the different pricing schemes that are offered by Amazon RDS:

- **Pay-as-you-go**: This is a flexible pricing model that does not involve any upfront costs, and you only pay for the resources you use.
- **Reserved instances**: You can save money with reserved instances by committing to use RDS for a 1- or 3-year term.
- **Free tier**: New users can get started with Amazon RDS for free, with limited usage included under the AWS Free Tier.
- **Cost management tools**: AWS Cost Explorer and Trusted Advisor can be used to monitor usage and optimise costs.

3.5 Security and Compliance

- **Encryption**: This enables data encryption at rest and in transit to protect sensitive information.
- **VPC integration**: Run your database instances in an Amazon Virtual Private Cloud (VPC) for network isolation.
- **IAM integration**: Use AWS Identity and Access Management (IAM) to control access to your RDS instances.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Compliance certifications**: Amazon RDS is compliant with various industry standards, including HIPAA, PCI DSS and SOC.

3.6 Performance and Scalability

- **Instance types**: Choose from a variety of instance types optimised for different use cases and performance needs, including general-purpose, memory-optimised and burstable performance instances.
- **Storage options**: Select from multiple storage options, such as General Purpose SSD, Provisioned IOPS SSD and Magnetic Storage, to balance performance and cost.
- **Auto scaling**: Automatically adjust storage capacity and compute resources to meet changing workload demands without downtime.
- **Read replicas**: Enhance read performance and scalability by creating read replicas in different regions.

3.7 Management and Administration

- **AWS Management Console**: Easily manage your RDS instances through the AWS Management Console.
- **CLI and SDKs**: Use the AWS Command Line Interface (CLI) and Software Development Kits (SDKs) for scripting and automation.
- **Monitoring and alerts**: Utilise Amazon CloudWatch to set up monitoring and alerts for your database instances.

3.8 Backup and Recovery

- **Automated backups**: Enable automated backups for daily snapshots and point-in-time recovery.
- **Manual snapshots**: Create and manage manual snapshots for additional backup control.
- **Cross-region backup**: Copy snapshots to different AWS regions for disaster recovery.

3.9 Migration and Integration

- **Database Migration Service (DMS)**: Simplify database migrations to Amazon RDS with minimal downtime using AWS DMS.
- **Schema Conversion Tool (SCT)**: Convert database schemas from different database engines to Amazon RDS.
- **Integration with AWS Services**: Seamlessly integrate RDS with other AWS services such as Lambda, S3 and CloudFormation.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

3.10 Best Practices

- **Instance selection**: Choose the right instance type and storage option based on your workload requirements.
- **Security measures**: Implement security best practices, including encryption, IAM policies and VPC configurations.
- **Performance tuning**: Regularly monitor and tune your database performance using Amazon RDS performance insights and CloudWatch.

4. Amazon Aurora

4.1 Introduction to Amazon Aurora

Amazon Aurora is a high-performance, MySQL- and PostgreSQL-compatible relational database service designed for the cloud. It combines the speed and reliability of high-end commercial databases with the simplicity and cost-effectiveness of open-source databases. Aurora is designed to provide up to five times the performance of MySQL and up to three times the performance of PostgreSQL, with enhanced scalability and availability.

4.2 Key Features

- **High performance**: Delivers exceptional throughput and low latency for demanding applications
- **Compatibility**: Fully compatible with MySQL and PostgreSQL, enabling easy migration and integration
- **Auto-scaling storage**: Automatically scales storage up to 128 TB as your database grows
- **Replication**: Supports up to 15 low-latency read replicas for improved read performance
- **Global databases**: Allows you to replicate your data across multiple AWS regions for global availability

4.3 Performance and Scalability

- **Cluster architecture**: Designed with a distributed, fault-tolerant, self-healing storage system that scales automatically
- **Read and write performance**: Optimised for fast read and write operations, providing high throughput and low latency
- **Compute scaling**: Easily scales compute resources up or down to handle varying workloads

4.4 High Availability and Durability

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Multi-AZ deployments**: It ensures high availability with automated failover across multiple Availability Zones.
- **Fault-tolerant storage**: Data is automatically replicated across multiple disks in multiple Availability Zones.
- **Continuous backups**: It enables continuous backup to Amazon S3 with point-in-time recovery.

4.5 Supported Database Engines

- **MySQL-compatible**: Amazon Aurora provides compatibility with MySQL, delivering enhanced performance and availability while maintaining MySQL's familiar interface and tools.
- **PostgreSQL-Compatible**: Amazon Aurora also offers compatibility with PostgreSQL, combining PostgreSQL's advanced features with Aurora's scalability and durability.

4.6 Backup and Recovery

- **Automated backups**: Enable automated backups for daily snapshots and point-in-time recovery.
- **Manual snapshots**: Create and manage manual snapshots for additional backup control.
- **Point-in-time recovery**: Restore your database to any specific time within your backup retention period.

4.7 Best Practices

- **Instance selection**: Choose the right instance type and storage option based on your workload requirements.
- **Security measures**: Implement security best practices, including encryption, IAM policies and VPC configurations.
- **Performance tuning**: Regularly monitor and tune your database performance using Amazon Aurora performance insights and CloudWatch.

5. Amazon Redshift

5.1 Introduction to Amazon Redshift

Amazon Redshift is a fully managed data warehouse service designed for large-scale data analysis and complex queries. It enables businesses to analyse their data using SQL and existing business intelligence tools, offering fast query performance and scalability to petabyte-scale data.

5.2 Key Features

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Massively Parallel Processing (MPP)**: Distributes queries across multiple nodes for fast, efficient processing
- **Columnar storage**: Stores data by columns, reducing I/O and improving query performance
- **Data compression**: Automatically compresses data, saving storage and enhancing performance
- **Elastic resize**: Quickly resizes clusters to meet workload demands

5.3 Performance and Scalability

- **Scalable architecture**: Easily scales from a few hundred gigabytes to a petabyte or more
- **Concurrency scaling**: Automatically adds capacity to handle sudden increases in the query demand
- **High-performance query engine**: Is optimised for complex queries and large data sets

5.4 High Availability and Durability

- **Automated backups**: Redshift continuously backs up data to Amazon S3.
- **Replication**: Data is automatically replicated within the cluster and across Availability Zones.
- **Cluster recovery**: It involves automated recovery processes to handle hardware failures.

5.5 Data Warehousing Architecture

- **Distributed storage**: Data is distributed across multiple nodes for parallel processing.
- **Advanced indexing**: It supports efficient data retrieval and complex queries.
- **Materialised views**: It provides precomputed views for faster query performance.

5.6 Query and Data Management

- **SQL support**: It provides comprehensive support for SQL queries, including joins, aggregations and window functions
- **Data import/export**: It can easily load data from S3, DynamoDB, and other data sources and export query results back to S3.
- **Data transformation**: You can use SQL-based ETL processes to transform and prepare data for analysis.

5.7 Integration With BI Tools

- **Compatibility**: It seamlessly integrates with popular BI tools such as Tableau, Looker and Amazon QuickSight

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Data visualisation**: It can leverage BI tools to create interactive dashboards and visualisations.

5.8 Backup and Recovery

- **Automated backups**: Enable automated backups for daily snapshots and point-in-time recovery.
- **Manual snapshots**: Create and manage manual snapshots for additional backup control.
- **Cross-region backup**: Copy snapshots to different AWS regions for disaster recovery.

5.9 Best Practices

- **Instance selection**: Choose the right instance type and storage option based on your workload requirements.
- **Security measures**: Implement security best practices, including encryption, IAM policies and VPC configurations.
- **Performance tuning**: Regularly monitor and tune your database performance using Amazon Redshift performance insights and CloudWatch.

6. Amazon Neptune

6.1 Introduction to Amazon Neptune

Amazon Neptune is a fully managed graph database service designed to build and run applications that work with highly connected data sets. It supports the property graph and RDF graph models, enabling efficient querying of complex, interconnected data.

6.2 Key Features

- **Multi-model support**: Supports the property graph and RDF graph models
- **ACID compliance**: Ensures transactional consistency and reliability
- **Low latency**: Is optimised for fast graph queries and traversal
- **Fully managed**: Automates backups, patching and scaling

6.3 Performance and Scalability

- **Horizontal scaling**: It allows for the scaling of read capacity with up to 15 low-latency read replicas.
- **High throughput**: It is designed to handle large volumes of concurrent graph queries efficiently.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Efficient storage**: It features an optimized storage engine specifically for graph data structures.

6.4 High Availability and Durability

- **Multi-AZ deployments**: It ensures high availability with automated failover across multiple Availability Zones.
- **Replication**: Data is automatically replicated across multiple disks and Availability Zones.
- **Cluster recovery**: It offers automated recovery processes to handle hardware failures.

6.5 Graph Database Architecture

- **Property graph model**: You can use Gremlin to traverse property graphs with the property graph model.
- **RDF graph model**: For querying RDF graphs, use the RDF graph model with SPARQL.
- **Efficient indexing**: Efficient indexing is optimized for fast graph traversal and query performance.

6.6 Query and Data Management

- **Gremlin support**: Execute complex traversals and queries using Apache TinkerPop's Gremlin.
- **SPARQL support**: Perform expressive queries over RDF data using SPARQL.
- **Data import/export**: Easily import and export data from various sources.

6.7 Integration With Analytics and Visualisation Tools

- **Compatibility**: Integrates with popular analytics and visualisation tools such as Amazon QuickSight and Neo4j.
- **Data visualisation**: Create interactive graphs and visualisations to analyse relationships in data.

6.8 Backup and Recovery

- **Automated backups**: Enable automated backups for daily snapshots and point-in-time recovery.
- **Manual snapshots**: Create and manage manual snapshots for additional backup control.
- **Cross-region backup**: Copy snapshots to different AWS regions for disaster recovery.

6.9 Best Practices

- **Instance selection**: Choose the right instance type and storage option based on your workload requirements.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Security measures**: Implement security best practices, including encryption, IAM policies and VPC configurations.
- **Performance tuning**: Regularly monitor and tune your database performance using Amazon Neptune performance insights and CloudWatch.

7. Amazon Neptune

7.1 Introduction to Amazon DocumentDB

Amazon DocumentDB is a fully managed document database service designed to be compatible with MongoDB. It is optimised for JSON data, providing scalability, high availability and performance, making it an ideal solution for modern applications that need flexible and efficient database services.

7.2 Key Features

- **MongoDB compatibility**: Seamlessly runs and manages MongoDB workloads
- **Fully managed**: Automates database management tasks such as backups, patching and scaling
- **Scalable**: Automatically scales storage and compute capacity
- **High availability**: Ensures data availability across multiple Availability Zones

7.3 Performance and Scalability

- **Auto-scaling storage**: Automatically scales storage from 10 GB to 64 TB
- **High throughput**: Optimised for high performance and low latency
- **Read replicas**: Supports up to 15 read replicas for improved read scalability

7.4 High Availability and Durability

- **Multi-AZ deployments**: Ensure high availability with automated failover across multiple Availability Zones.
- **Replication**: Data is automatically replicated across multiple disks and Availability Zones.
- **Cluster recovery**: Automated recovery processes handle hardware failures.

7.5 MongoDB Compatibility

- **MongoDB APIs**: It is compatible with MongoDB 3.6 APIs, making it easy to migrate applications.
- **Development tools**: It supports MongoDB drivers and tools, allowing for seamless integration.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

7.6 Query and Indexing

- **Advanced query capabilities**: Supports complex queries and indexing
- **JSON support**: Natively supports JSON data structures
- **Aggregation framework**: Uses MongoDB's aggregation framework for data analysis

7.7 Backup and Recovery

- **Automated backups**: Enable automated backups for daily snapshots and point-in-time recovery.
- **Manual snapshots**: Create and manage manual snapshots for additional backup control.
- **Cross-region backup**: Copy snapshots to different AWS regions for disaster recovery.

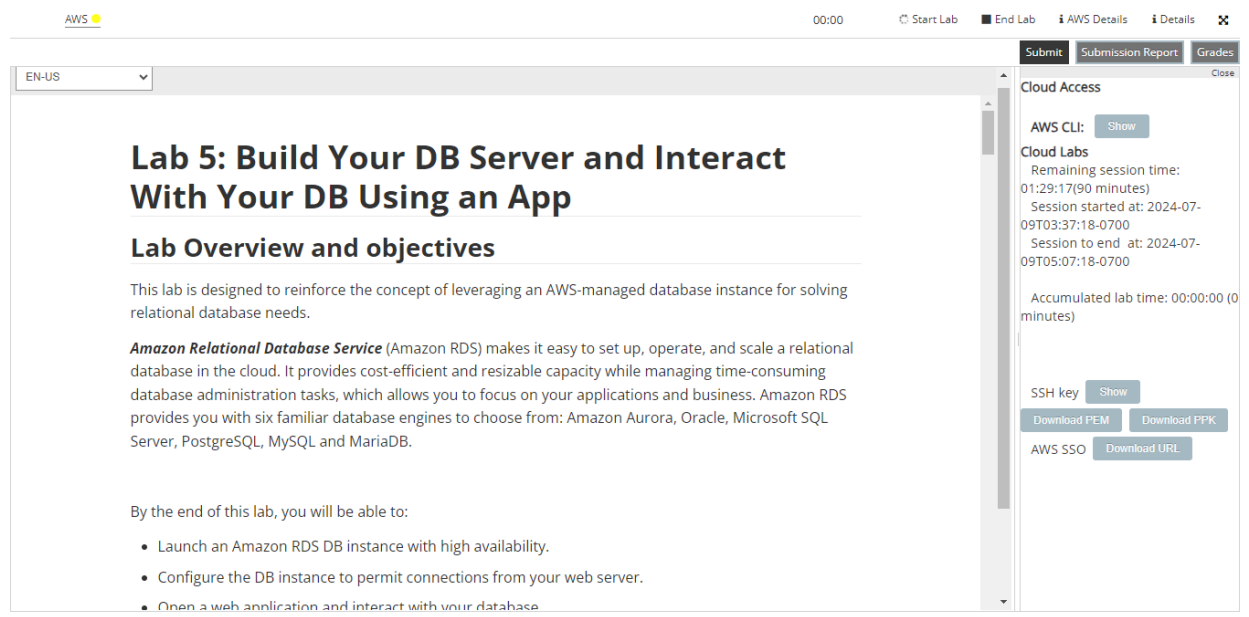
7.8 Best Practices

- **Instance selection**: Choose the right instance type and storage option based on your workload requirements.
- **Security measures**: Implement security best practices, including encryption, IAM policies and VPC configurations.
- **Performance tuning**: Regularly monitor and tune your database performance using Amazon DocumentDB performance insights and CloudWatch.

8. Hands-on/Lab Demo

1. At the top of the instructions mentioned there, select **Start Lab**.
    - The lab session will begin.
    - A timer is present at the top of the page, which displays the time remaining for the session to complete.
      Tip: If you need to refresh the session length at any given point, choose **Start Lab** again before the timer reaches 0:00.
    - Before continuing, wait until the circle icon present to the right of the AWS link in the top left corner turns green.

© Copyright upGrad Education Pvt. Ltd. All rights reserved



2. To access the AWS Management Console, click on the **AWS** link located in the top left corner. This action will open a new tab in your browser, directing you to the console.

   Tip: If a new tab does not appear, your browser may be blocking pop-ups. Typically, a message or icon indicating this will appear at the top of your browser. Click on this message or icon, and select **Allow pop-ups** to proceed.

3. For convenience, arrange the AWS Management Console tab next to this set of instructions. By viewing both tabs simultaneously, you can easily follow the steps of the lab.

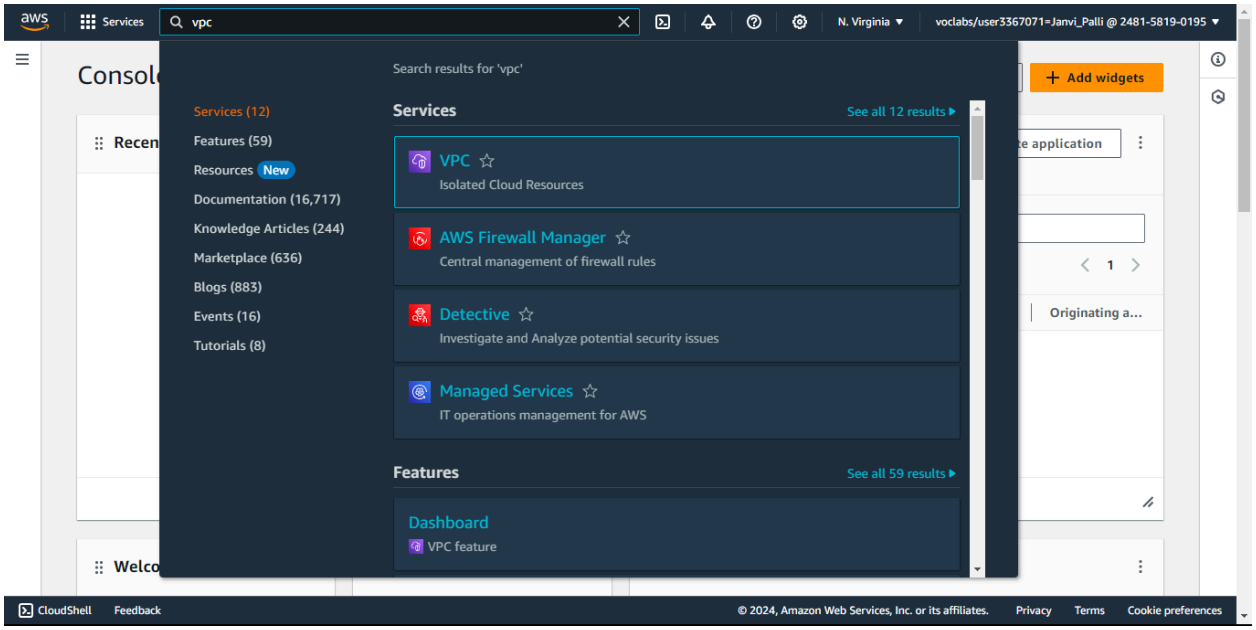


Task 1: Creating a Security Group for Your RDS DB Instance

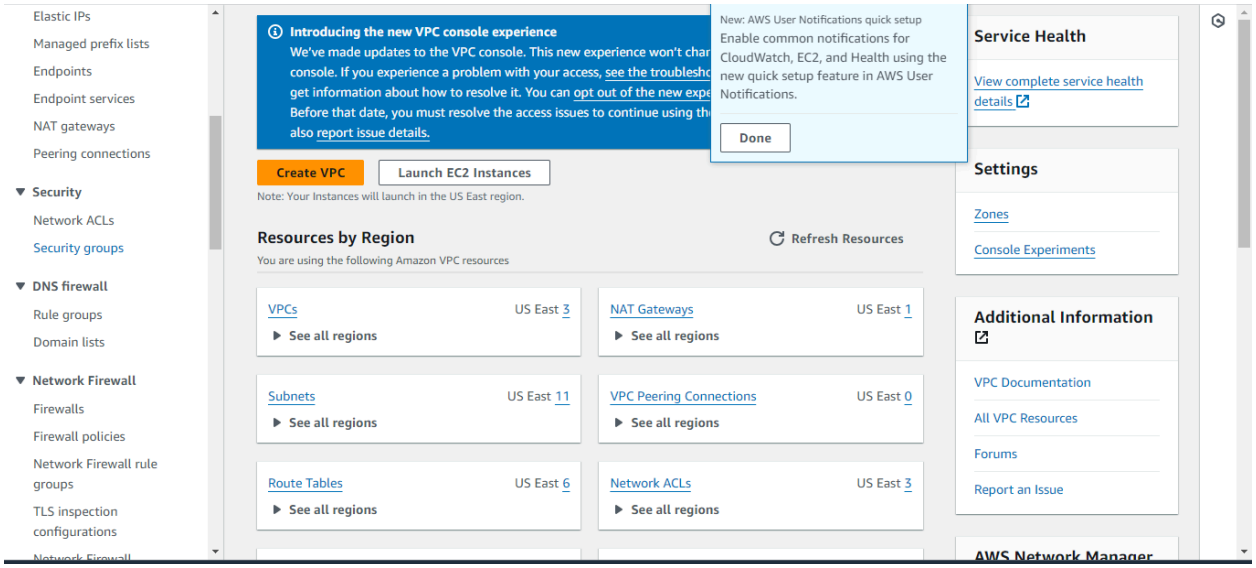
In this task, you will set up a security group to permit your web server to connect to your RDS database instance. This security group will be essential when initialising the database instance.

4. Begin by logging into the AWS Management Console. Locate the search box near the **Services** label, and enter **VPC** into it. Select **VPC** from the search results to proceed.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

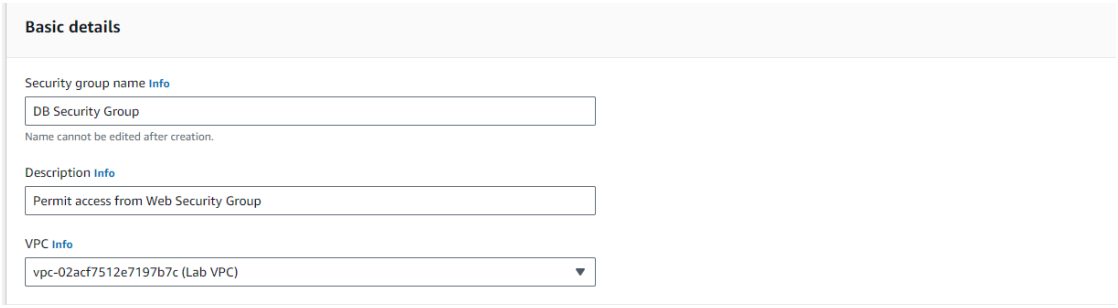


5. In the left navigation pane, choose **Security groups**.



6. Select **Create security group** to proceed. Then, configure the new security group with the following settings:
   - **Security group name:** DB Security Group
   - **Description:** Permit access from the Web Security Group
   - **VPC:** Select ‘Lab VPC’ by first removing any pre-selected VPC (click on the 'X' next to the current VPC) and then choosing ‘Lab VPC’ from the dropdown menu

© Copyright upGrad Education Pvt. Ltd. All rights reserved



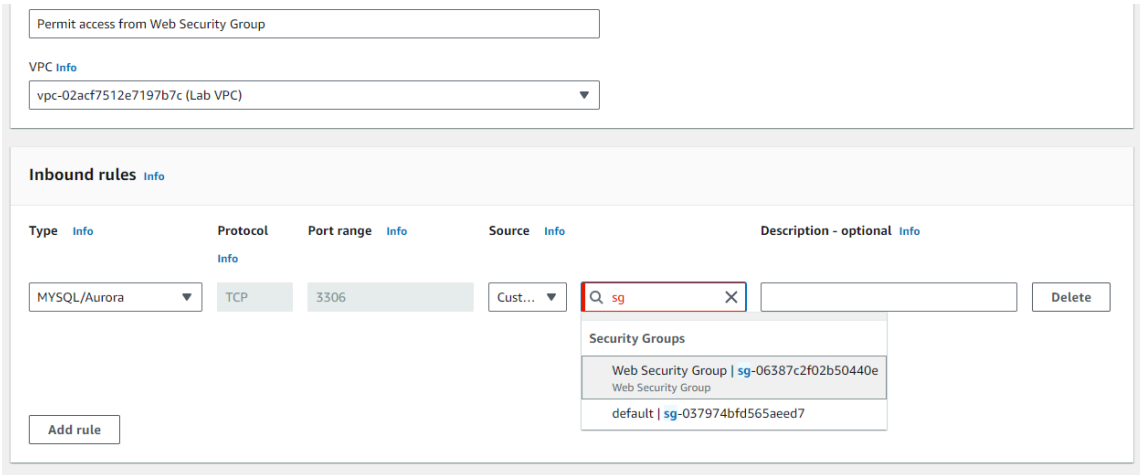
7. In the **Inbound rules** section of your new security group, click on **Add rule**.

Since no rules are currently active in this security group, you will need to add a rule to allow access from the Web Security Group.

8. Set up the rule with the following details:

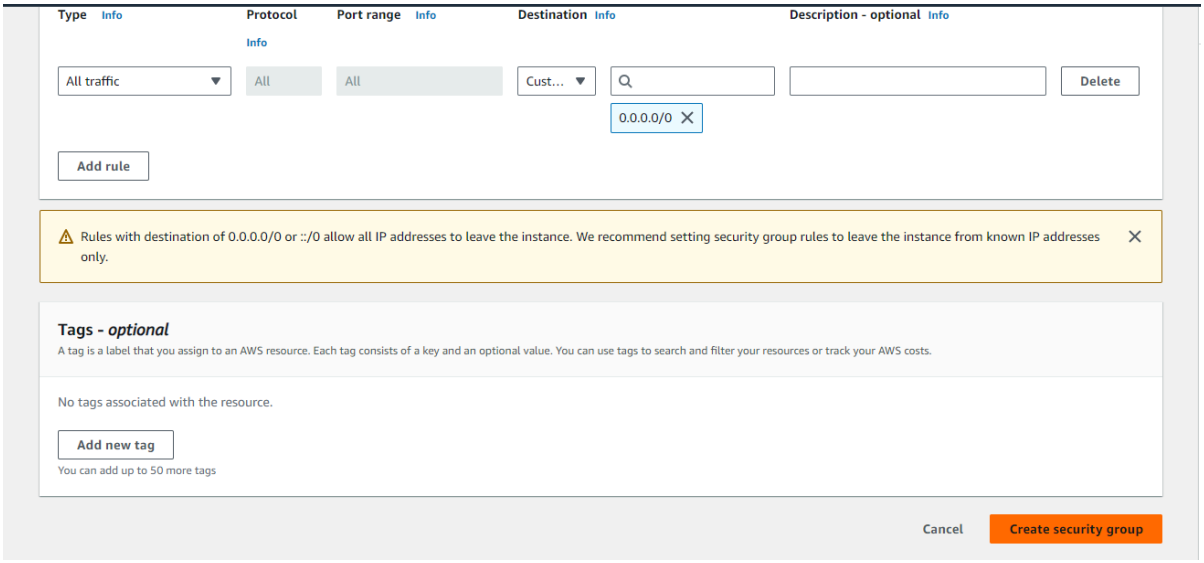
- **Type:** *MySQL/Aurora (3306)*
- **Source:** Move your cursor to the field right next to the **Custom** type **sg**, and select **Web Security Group** from the suggestions that appear.

This configuration enables the Database Security Group to allow inbound traffic on port 3306 from any EC2 instance associated with the Web Security Group.



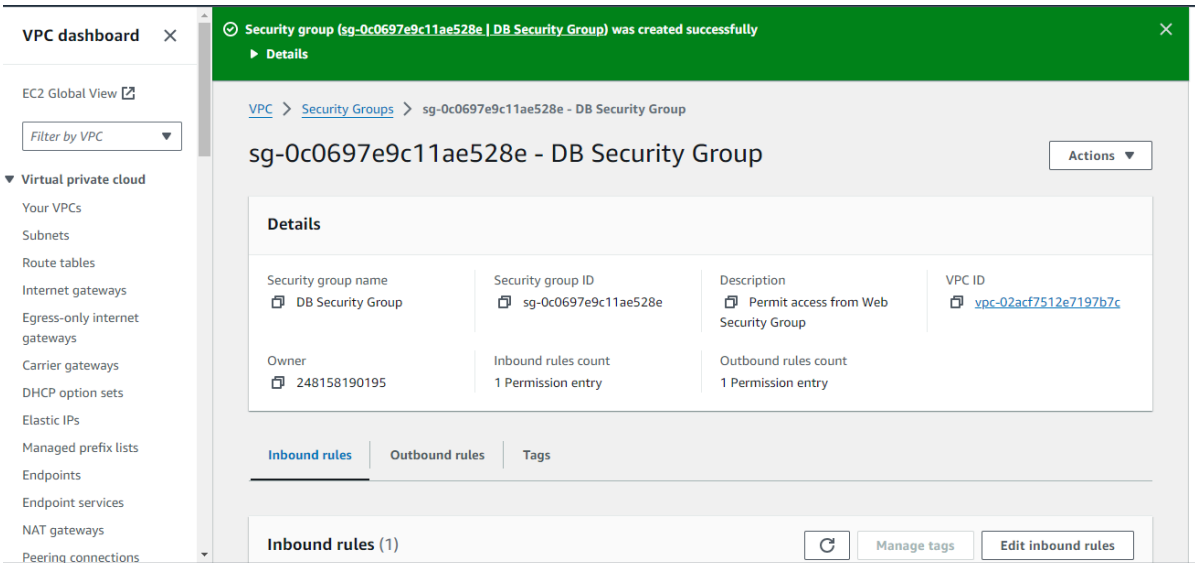
9. Select **Create security group**.
   This newly created security group will be used when you launch an Amazon RDS database in this lab.

© Copyright upGrad Education Pvt. Ltd. All rights reserved



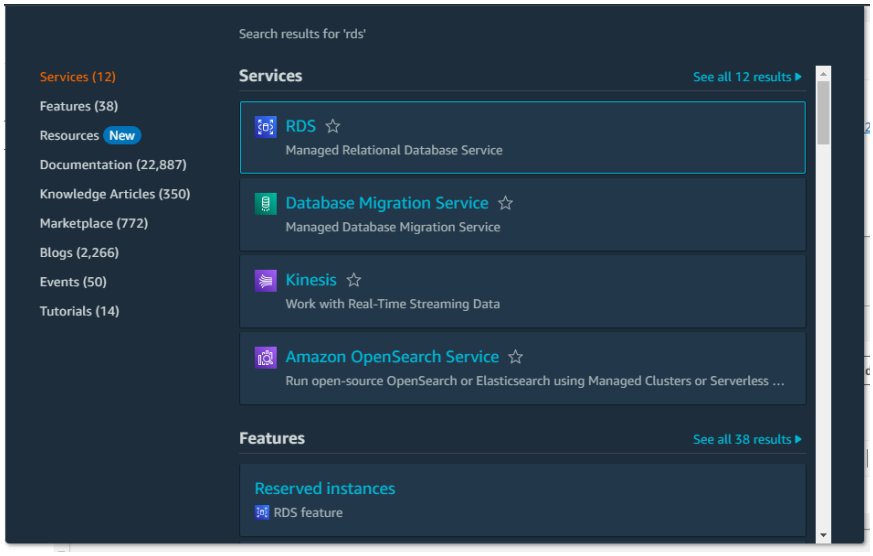
Task 2: Create a DB Subnet Group

In this task, you will create a **DB subnet group** that is used to instruct RDS which subnets can be used for the database. Each DB subnet group requires subnets in at least two Availability Zones.



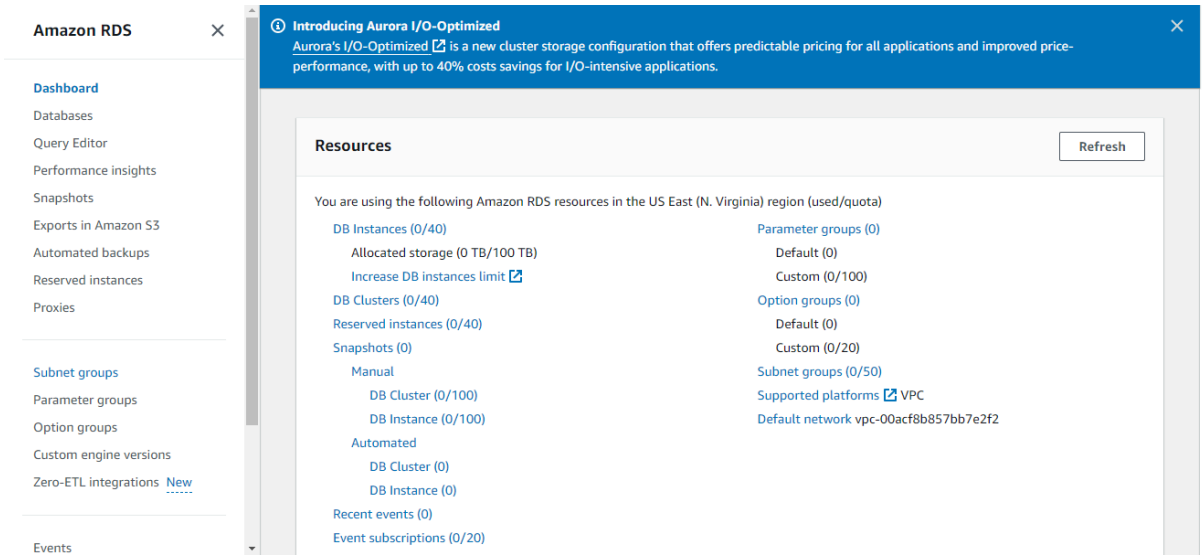
10. In the AWS Management Console, in the search box next to **Services**, search for and select **RDS**.

© Copyright upGrad Education Pvt. Ltd. All rights reserved



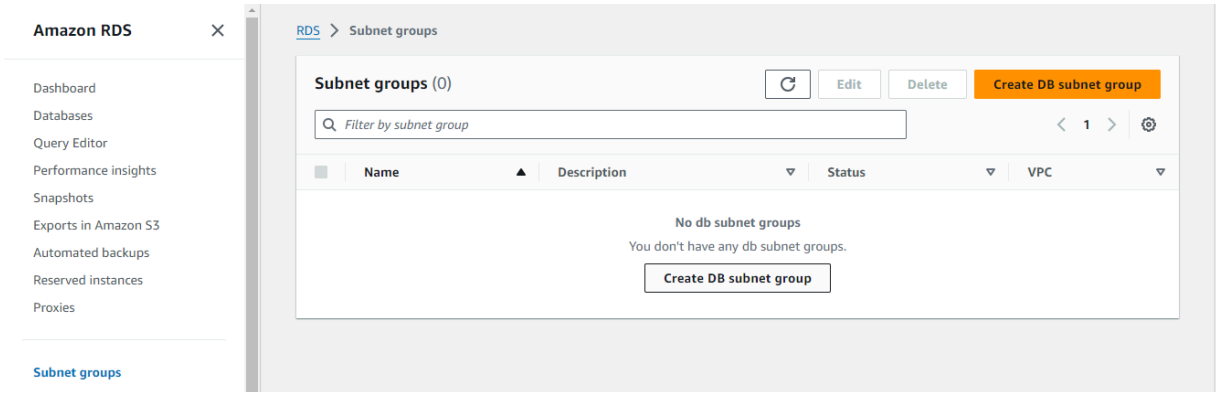
11. In the left navigation pane, choose **Subnet groups**.

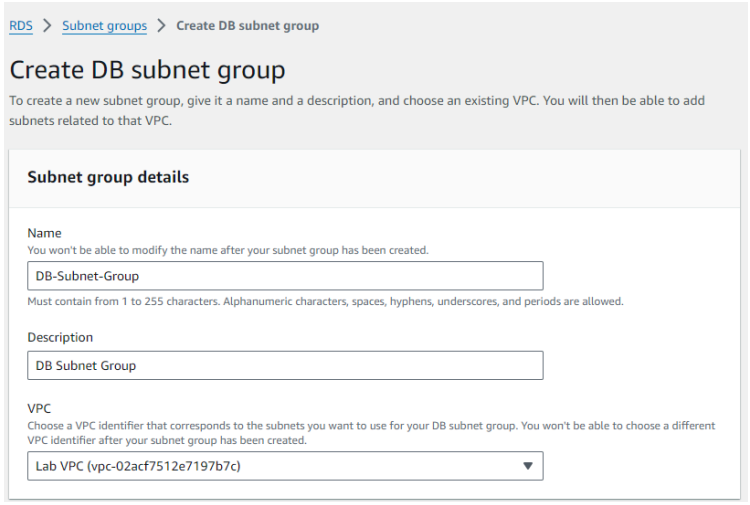
    If the navigation pane is not visible, choose the menu icon in the top left corner.



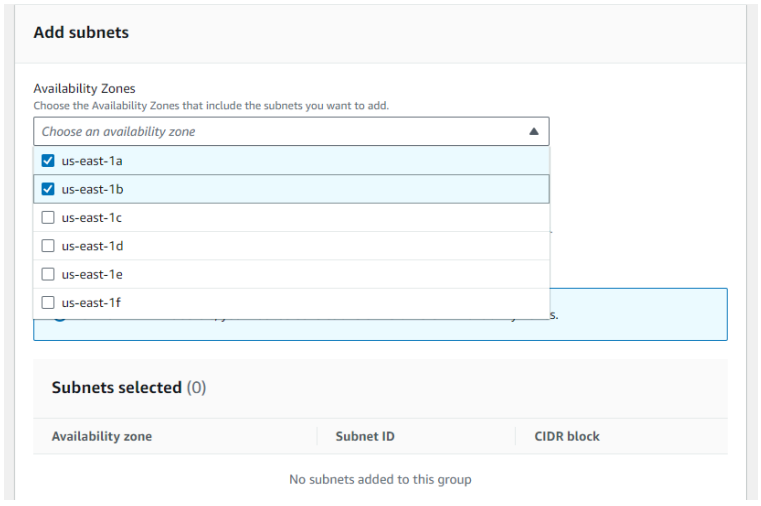
12. Choose **Create DB Subnet Group** and then configure the following settings:
    - **Name:** DB-Subnet-Group
    - **Description:** DB Subnet Group
    - **VPC:** *Lab VPC*

© Copyright upGrad Education Pvt. Ltd. All rights reserved



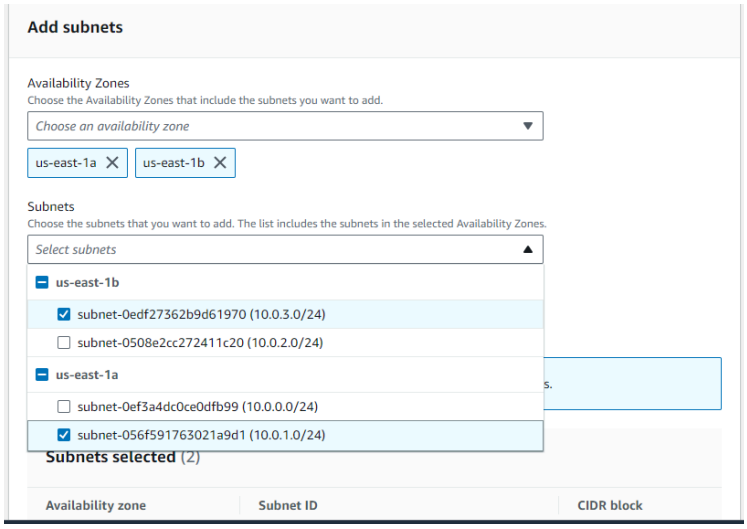


13. Scroll down to the **Add subnets** section.
14. Expand the list of values under **Availability Zones**, and select the first two zones - **us-east-1a** and **us-east-1b**.

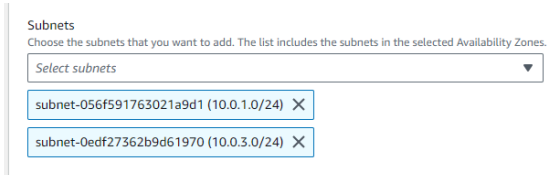


15. Expand the list of values under **Subnets**, and select the subnets associated with the CIDR ranges **10.0.1.0/24** and **10.0.3.0/24**.

© Copyright upGrad Education Pvt. Ltd. All rights reserved

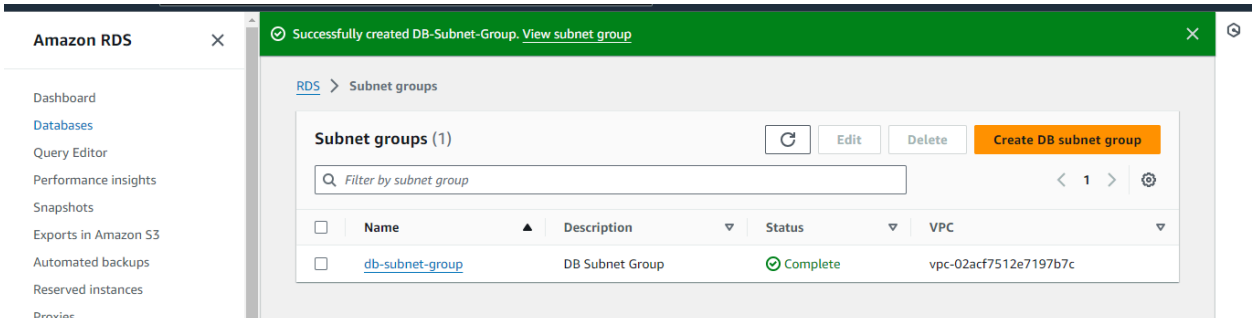


These subnets should now be shown in the **Subnets selected** table.



16. Select **Create**.

    You will use this DB subnet group when creating the database in the next task.



Task 3: Create an Amazon RDS DB Instance

In this task, you will configure and launch a Multi-AZ Amazon RDS deployment of a MySQL database instance.

Amazon RDS **Multi-AZ** deployments provide enhanced availability and durability for Database (DB) instances, making them a natural fit for production database workloads. When you

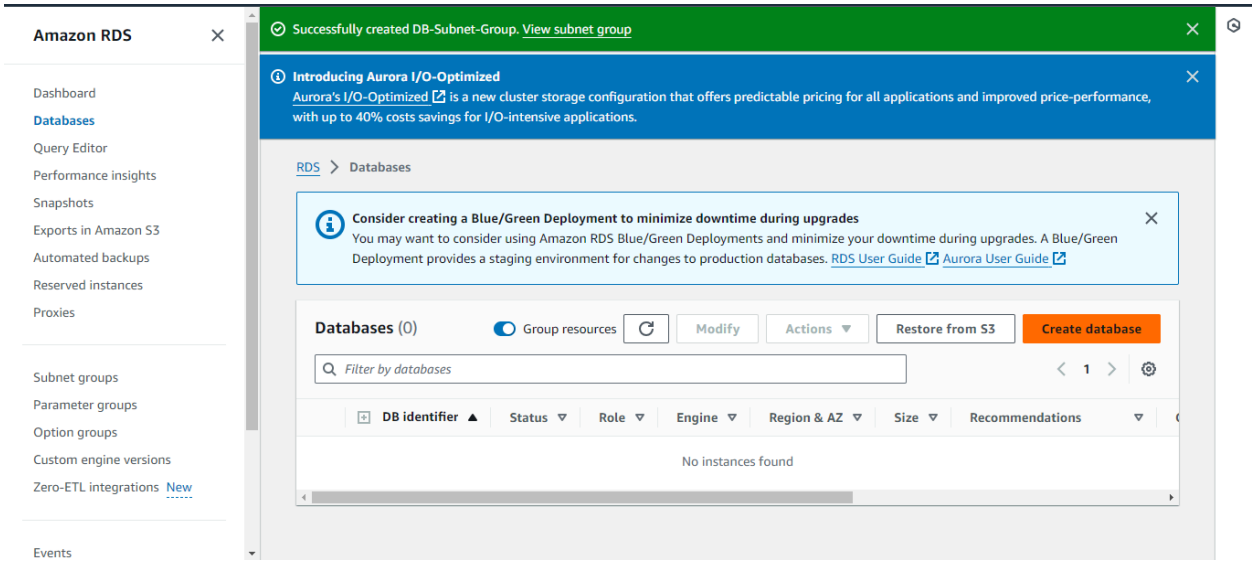
© Copyright upGrad Education Pvt. Ltd. All rights reserved

provision a Multi-AZ DB instance, Amazon RDS automatically creates a primary DB instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ).

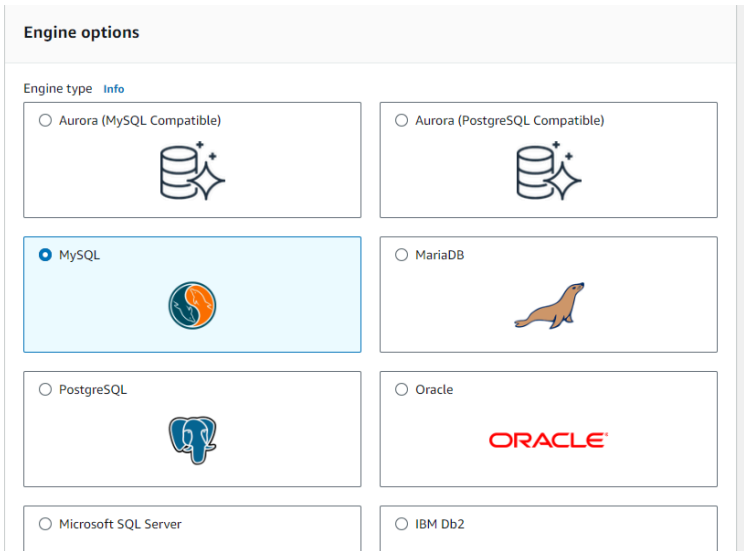
17. In the left navigation pane, choose **Databases**.

18. Select **Create database.**

Once you see **Switch to the new database creation flow** at the top of the screen, please choose it.

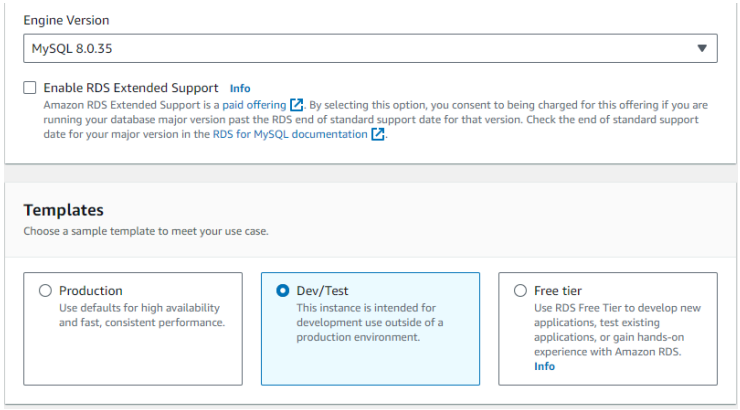


19. Select **MySQL** under **Engine Options**.

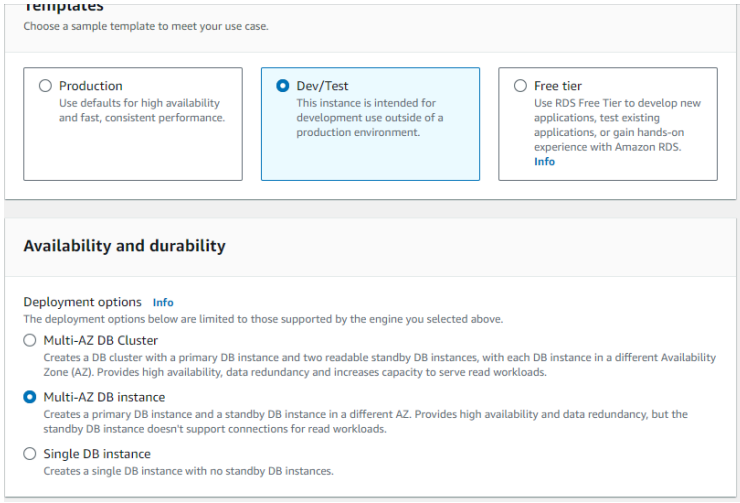


© Copyright upGrad Education Pvt. Ltd. All rights reserved

20. Under **Templates**, choose **Dev/Test**.

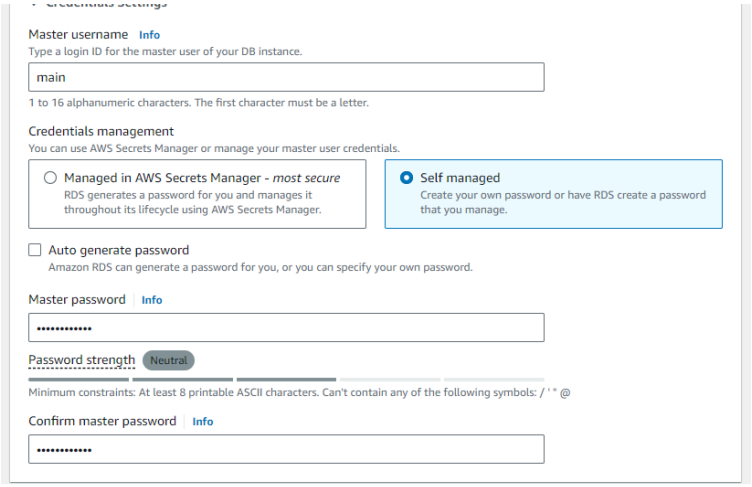


21. Under **Availability and durability**, select **Multi-AZ DB instance**.

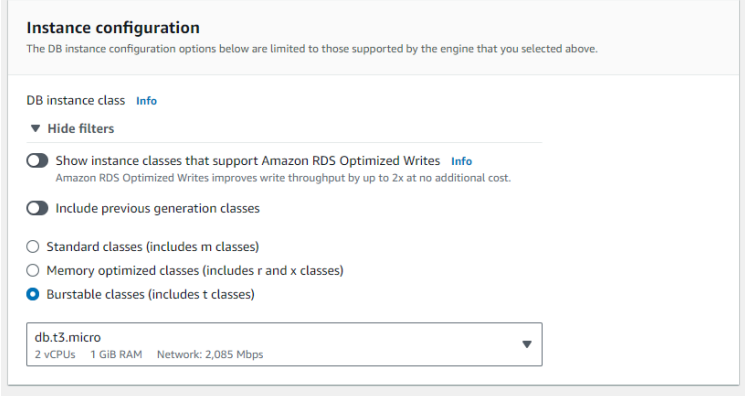


22. Under **Settings**, configure:
    - **DB instance identifier:** lab-db
    - **Master username:** main
    - **Master password:** lab-password
    - **Confirm password:** lab-password

© Copyright upGrad Education Pvt. Ltd. All rights reserved

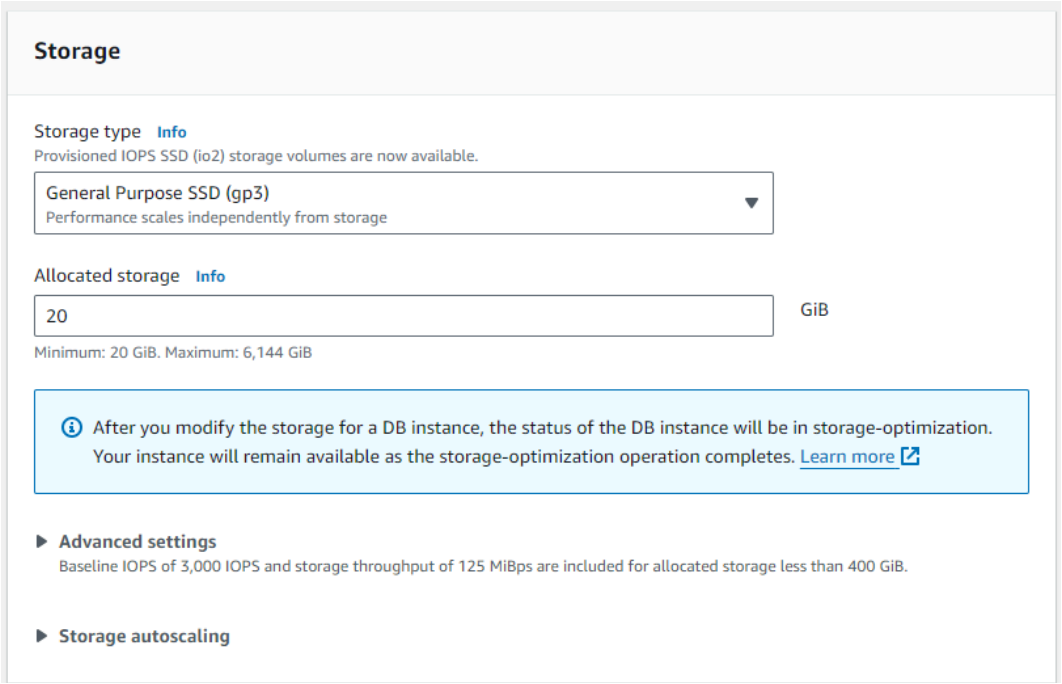


23. Under the **DB instance class**, configure:
    - Select **Burstable classes (includes t classes)**.
    - Select *db.t3.micro*.

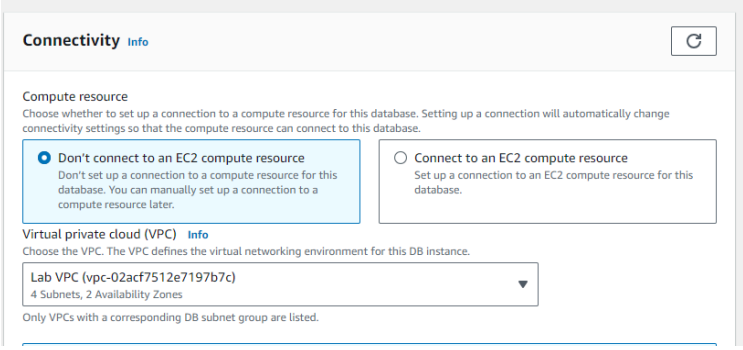


24. Under **Storage**, configure:
    - **Storage type:** *General Purpose (SSD)*
    - **Allocated storage:** *20*

© Copyright upGrad Education Pvt. Ltd. All rights reserved

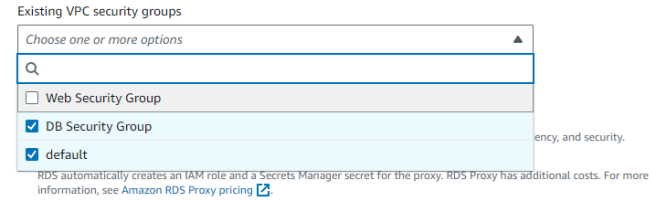


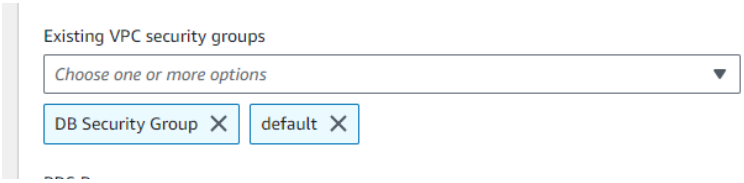
25. Under **Connectivity**, configure:
    - **Virtual Private Cloud (VPC):** *Lab VPC*



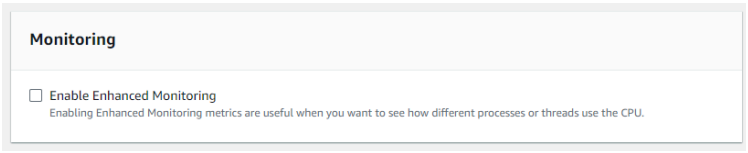
26. From the dropdown list under the **Existing VPC security groups**:
    - Choose *DB Security Group*.
    - Deselect *default*.

© Copyright upGrad Education Pvt. Ltd. All rights reserved



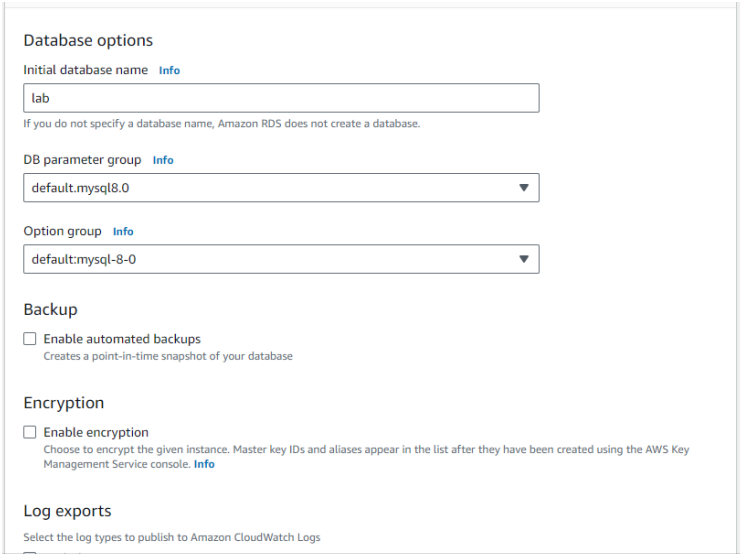


27. Under **Monitoring**, expand **Additional configuration**. Uncheck **Enable Enhanced monitoring**.



28. Under **Additional configuration**, configure:
    - **Initial database name:** lab
    - Uncheck **Enable automatic backups**
    - Uncheck **Enable encryption**

    This will turn off backups, which is not normally recommended, but it will make the database deploy faster for this lab.

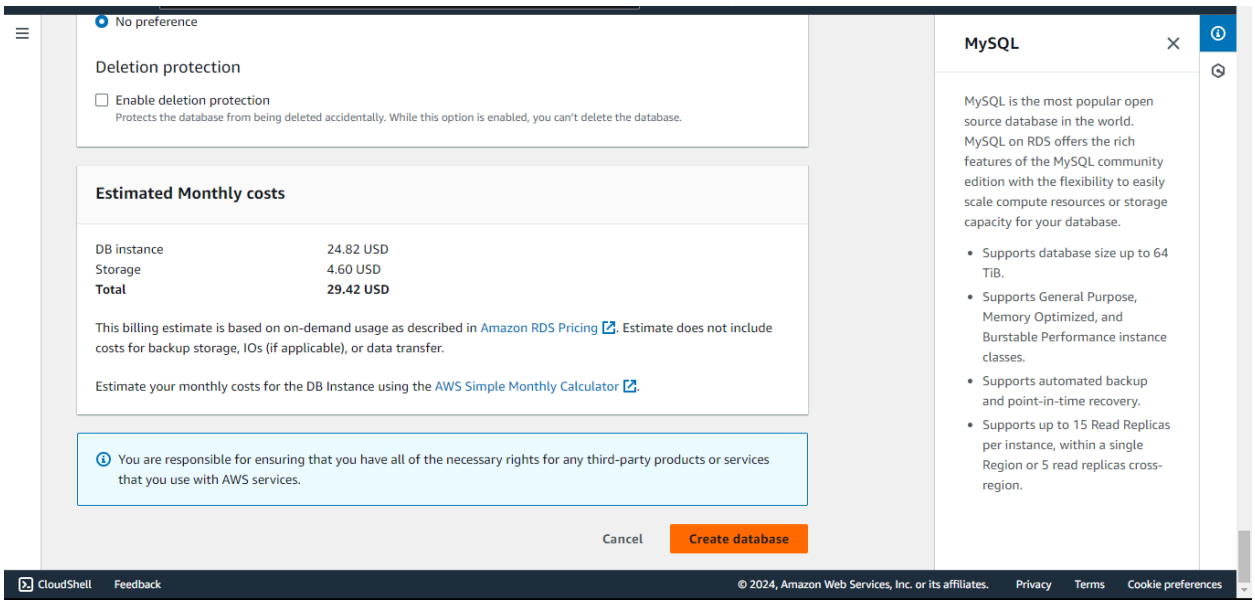


© Copyright upGrad Education Pvt. Ltd. All rights reserved

29. Select **Create database**.

Your database will now be launched.

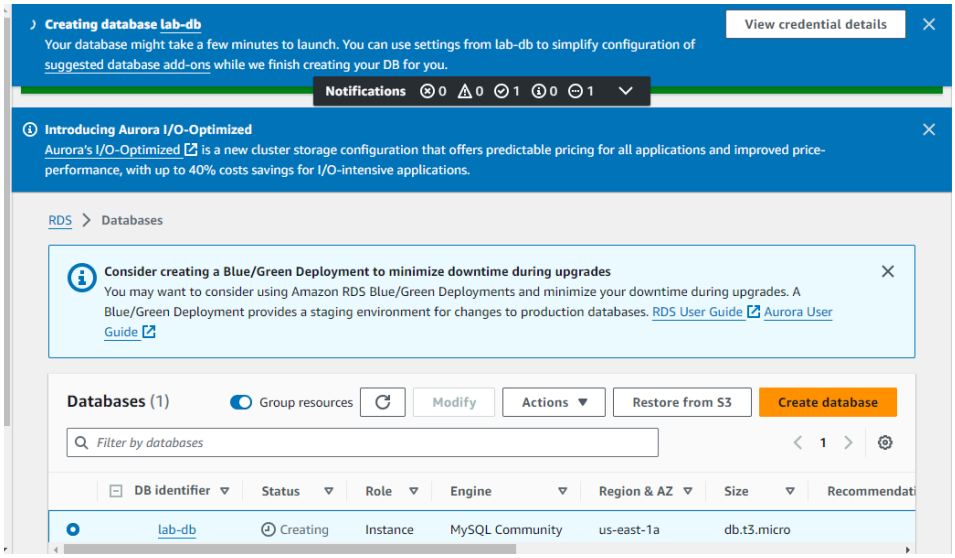
If you receive the error "not authorised to perform: iam:CreateRole," ensure you have unchecked **Enable Enhanced monitoring** in the previous step.



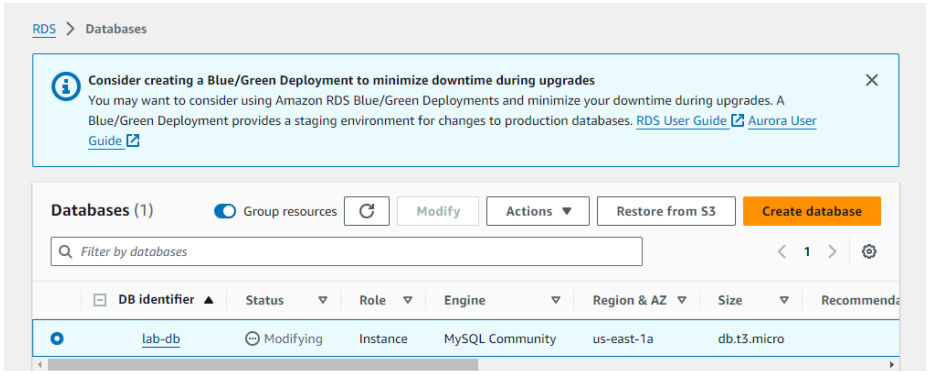
30. Choose **lab-db** (choose the link itself).

You will now need to wait **approximately 4 minutes** for the database to be available. The deployment process is now deploying a database in two different Availability Zones.

© Copyright upGrad Education Pvt. Ltd. All rights reserved



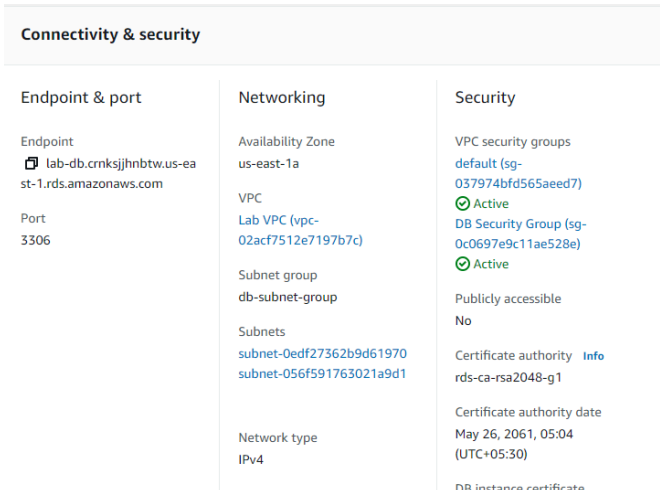
31. Wait until **Info** changes to **Modifying** or **Available**.



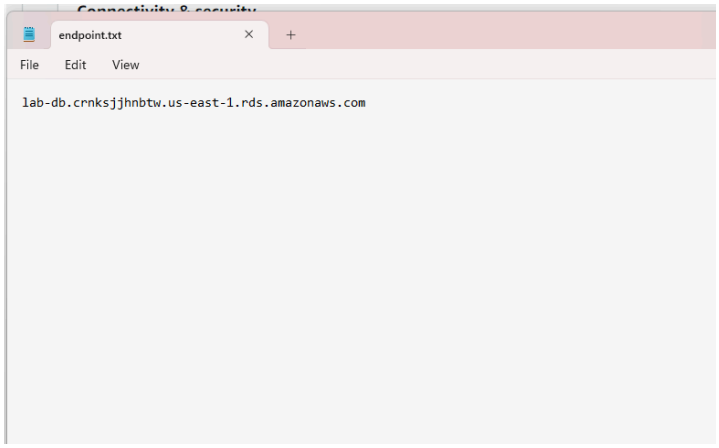
32. Scroll down to the **Connectivity & security** section, and copy the **Endpoint** field.

    It will look similar to *lab-db.xxxx.us-east-1.rds.amazonaws.com*.

© Copyright upGrad Education Pvt. Ltd. All rights reserved



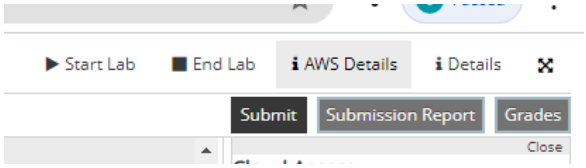
33. Paste the Endpoint value into a text editor. You will use it later in the lab.



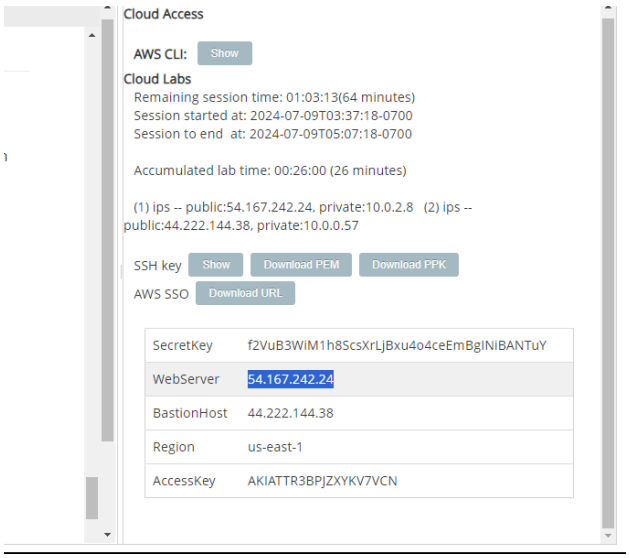
Task 4: Interact With Your Database

In this task, you will open a web application running on a web server that has been created for you. You will configure it to use the database you just created.

34. To discover the **WebServer** IP address, choose in the AWS Details dropdown menu above these instructions. Copy the IP address value.



© Copyright upGrad Education Pvt. Ltd. All rights reserved

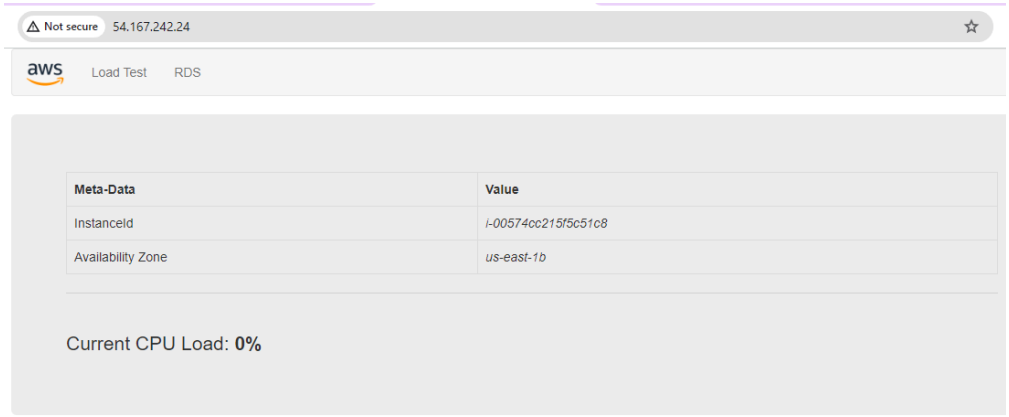


35. Open a new web browser tab, paste the WebServer IP address and press Enter.

    The web application will be displayed, showing information about the EC2 instance.

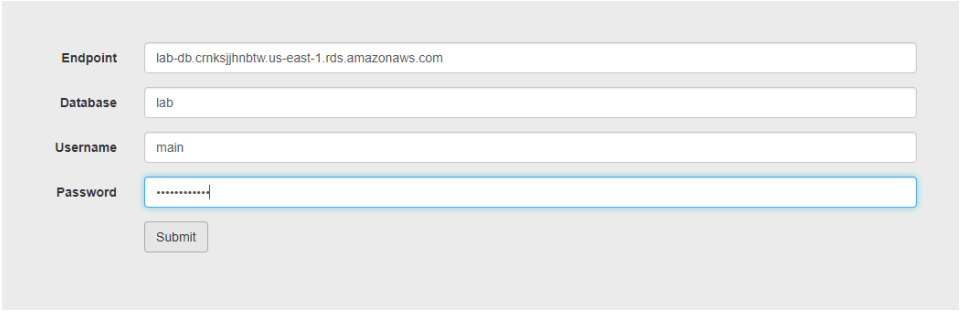
36. Choose the **RDS** link at the top of the page.

    You will now configure the application to connect to your database.

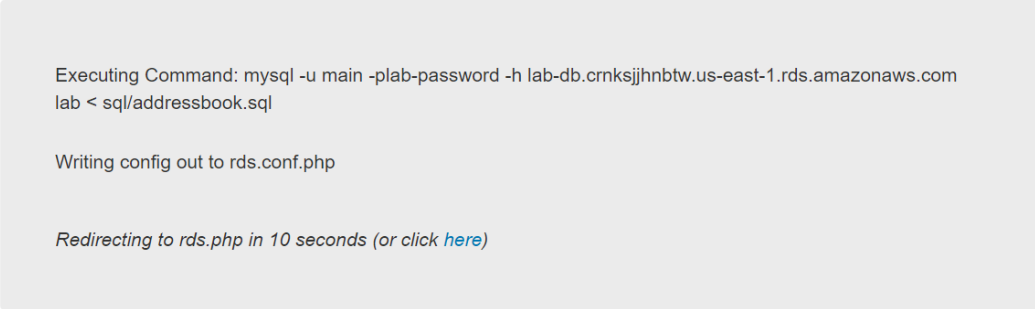


37. Configure the following settings:
    - **Endpoint:** Paste the Endpoint you copied to a text editor earlier
    - **Database:** lab
    - **Username:** main
    - **Password:** lab-password
    - Choose **Submit**

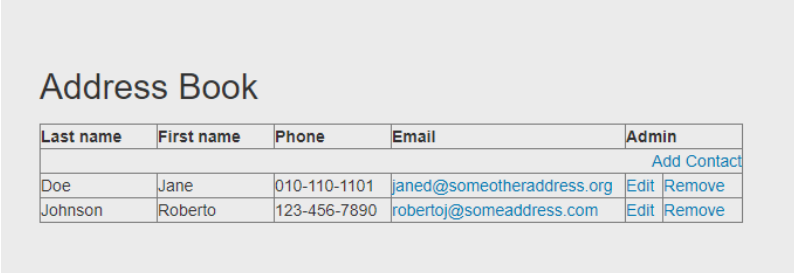
© Copyright upGrad Education Pvt. Ltd. All rights reserved



A message will appear explaining that the application is running a command to copy information to the database. After a few seconds, the application will display an **Address Book**.



The Address Book application will be using the RDS database to store information.



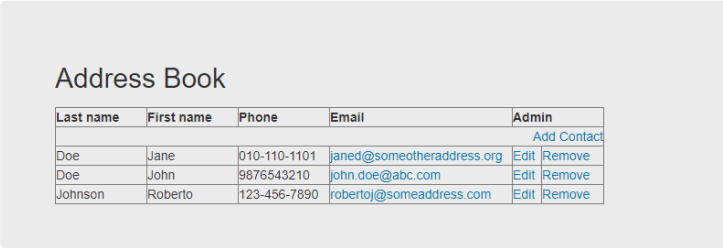
38. Test the web application by adding, editing and removing contacts.
    The data is persisted in the database and is automatically replicated to the second Availability Zone.



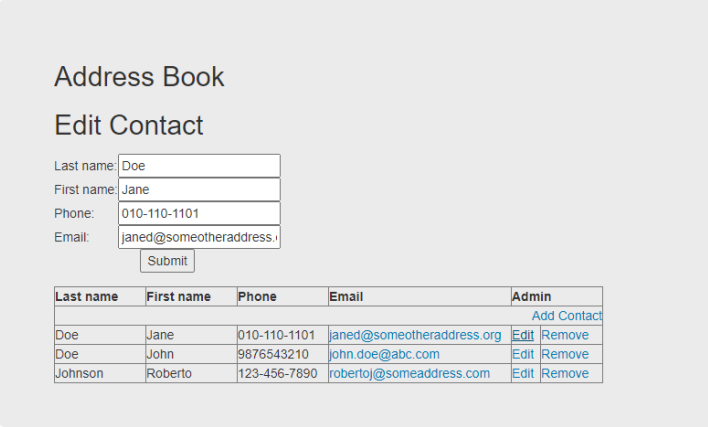
Adding a contact

© Copyright upGrad Education Pvt. Ltd. All rights reserved

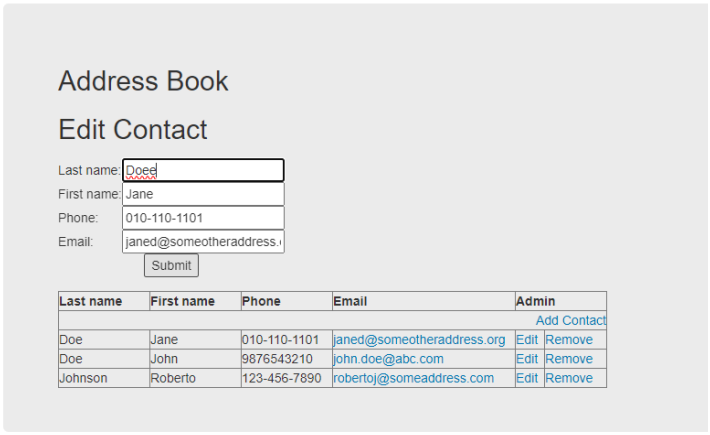


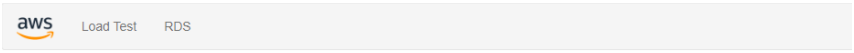


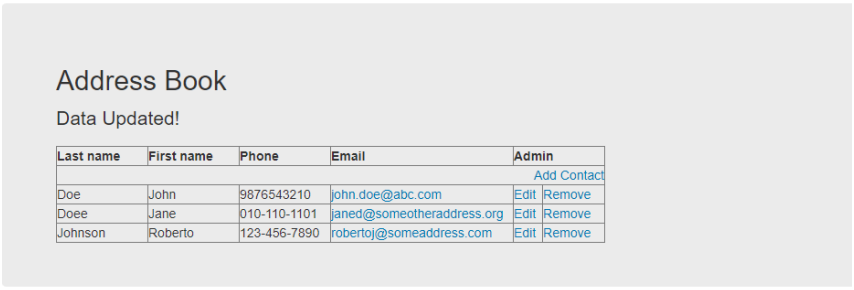
Editing a contact



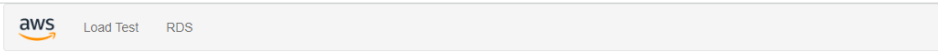
© Copyright upGrad Education Pvt. Ltd. All rights reserved



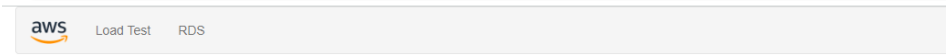


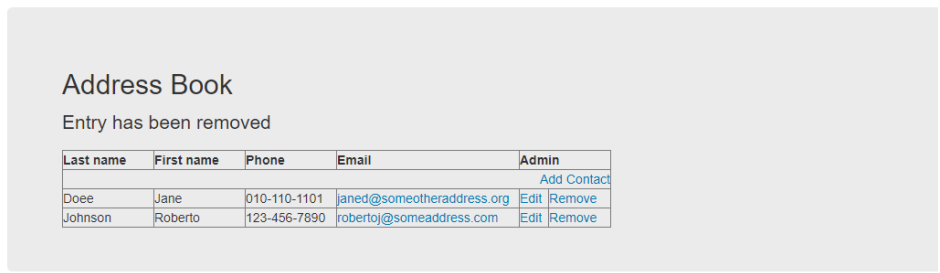


Removing a contact





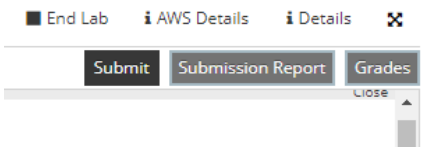




Submitting your work

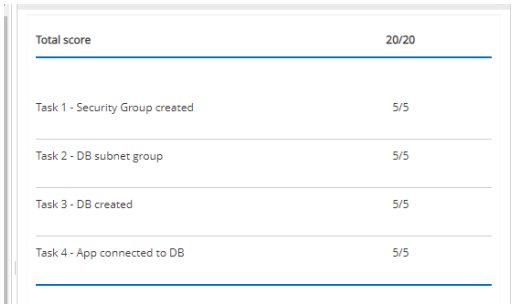
© Copyright upGrad Education Pvt. Ltd. All rights reserved

39. To record your progress, select **Submit** at the top of these instructions.

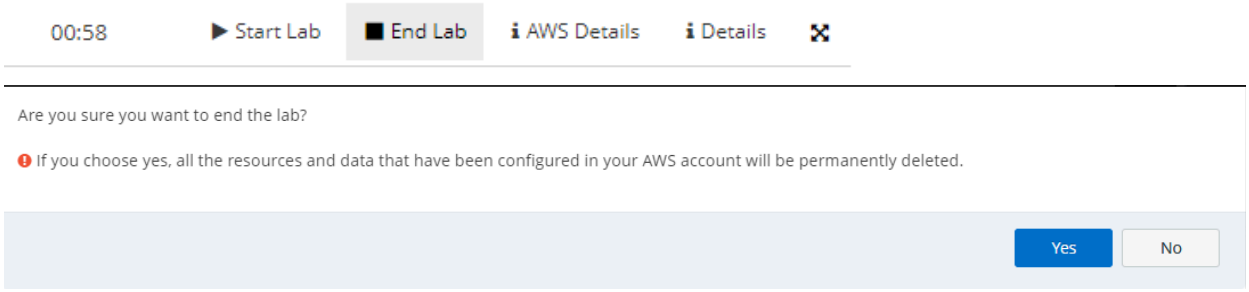


40. When prompted, choose **Yes**.

After a couple of minutes, the **grades** panel appears and shows the number of points you earned for each task. If the results are not displayed after a couple of minutes, select **Grades** at the top of these instructions.



41. Choose **End Lab** at the top of this page, and then choose **Yes** to confirm you want to end the lab.



9. Summary and Key Takeaways

Throughout this session, we have explored several crucial aspects of AWS databases. Here are the key takeaways:

Overview of AWS Databases

© Copyright upGrad Education Pvt. Ltd. All rights reserved

- **Definition and basic concepts**: AWS databases are managed services that handle various types of data, offering secure, scalable and reliable solutions.
- **Key characteristics of AWS databases**: These include automatic management of tasks, high performance, easy scaling, strong security and smooth integration with other AWS services.
- **Historical context and evolution of AWS databases**: AWS started with Amazon RDS and has since added many specialised databases such as DynamoDB and Neptune.

Types of AWS Database Services

- **Amazon RDS**: A managed relational database service that supports several database engines, including MySQL, PostgreSQL, Oracle, MariaDB and SQL Server
- **Amazon Aurora**: A high-performance database that is compatible with MySQL and PostgreSQL, offering automatic scaling and high availability
- **Amazon Redshift**: A fully managed data warehouse designed for large-scale data analysis
- **Amazon Neptune**: A managed graph database service that supports the property graph and RDF graph models
- **Amazon DynamoDB**: A managed NoSQL database that provides key-value and document data structures with very low latency
- **Amazon DocumentDB**: A managed document database service that is compatible with MongoDB workloads

Advantages and Disadvantages of AWS Databases

- Advantages
    - Managed services
    - Scalability
    - High performance
    - Security
    - Integration
- Disadvantages
    - Cost management
    - Vendor lock-in
    - Learning curve

By understanding these core concepts, types of services, benefits and challenges, you are now better equipped to make informed decisions about using AWS databases in your personal or professional environment.

© Copyright upGrad Education Pvt. Ltd. All rights reserved